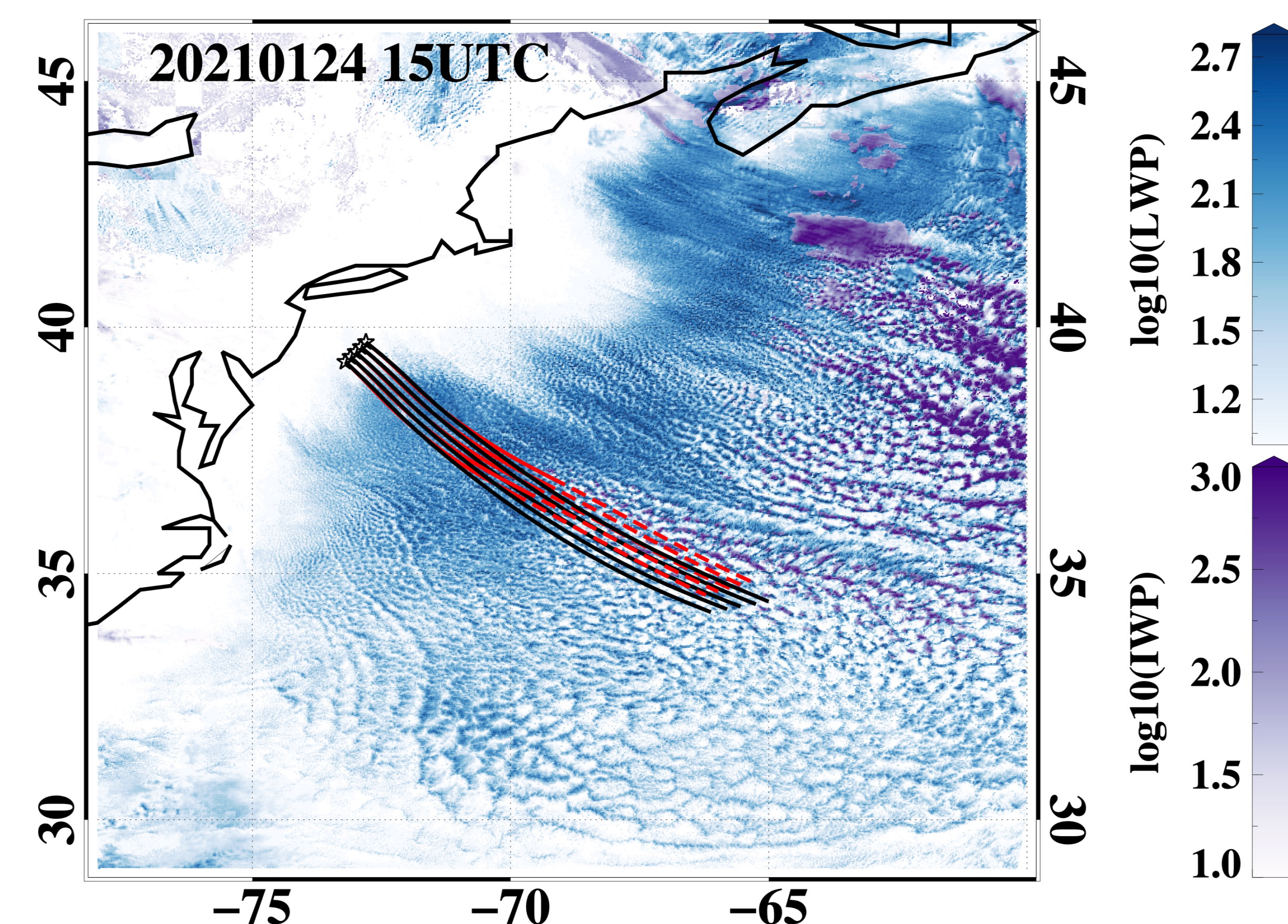


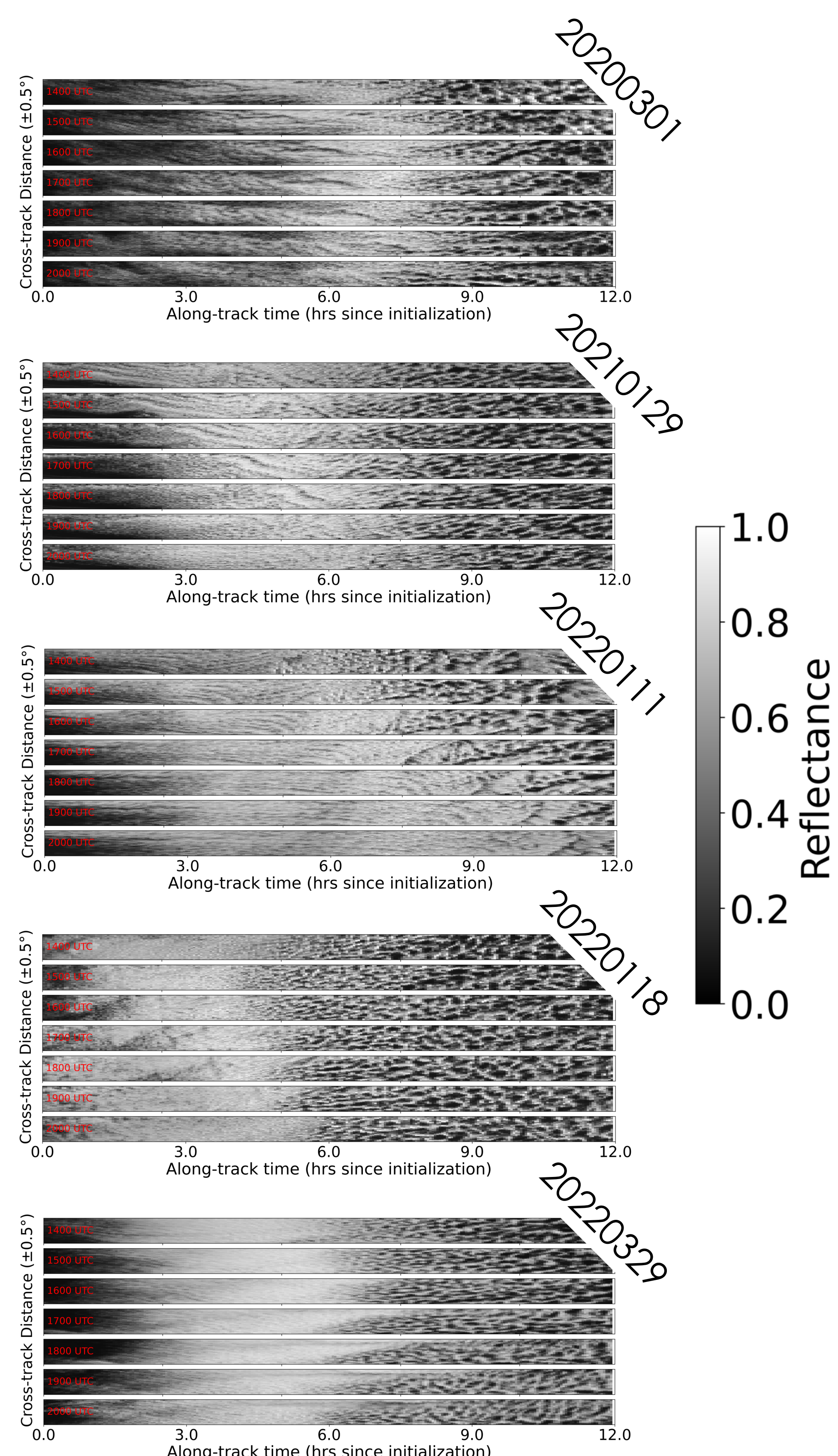


The 'time-space exchange'



- True Lagrangian trajectories (red)
 - solid: day time
 - dashed: night time
- Static "Lagrangian" trajectories (Black)

The five MCAO events (during NASA-ACTIVATE)

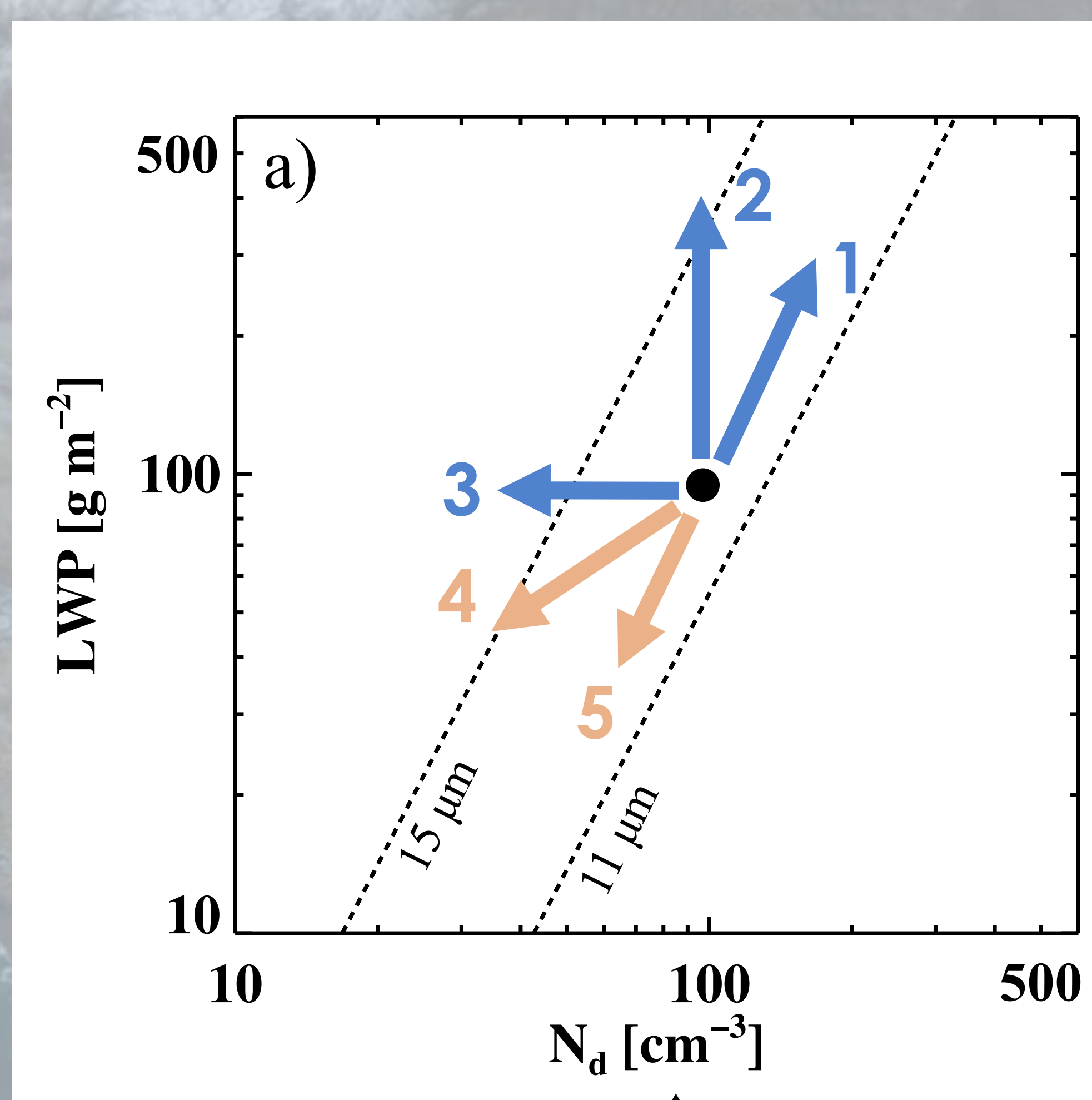


Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite

Jianhao Zhang^{1,2}, David Painemal³, Tom Dror^{1,2}, Jung-Sub Lim^{1,2}, Armin Sorooshian⁴, and Graham Feingold²

Takeaways:

- MCAO evolutions manifest in Micro-Macro physical spaces with **clear directionality**, indicating dominant cloud processes during morphological transitions.
- Evidence supports **two distinct groups** of mixed-phase processes: diffusional (slow) vs collisional (fast).
- Satellite snapshots can be used to reveal **characteristic fingerprints** that help infer cloud processes.



Fingerprints of Processes

1. Droplet activation
2. Condensational growth
3. Collision-coalescence
4. Precipitation
5. Entrainment (mixing and/or dilution)

Datasets:

GOES-16 cloud retrievals

- LWP, IWP, N_d , Albedo

ERA5 reanalysis

- Surface fluxes, M-index, RH700, ω_{700}

In-situ Data (NASA-ACTIVATE)

- LWC and IWC from FCDP and 2D-S

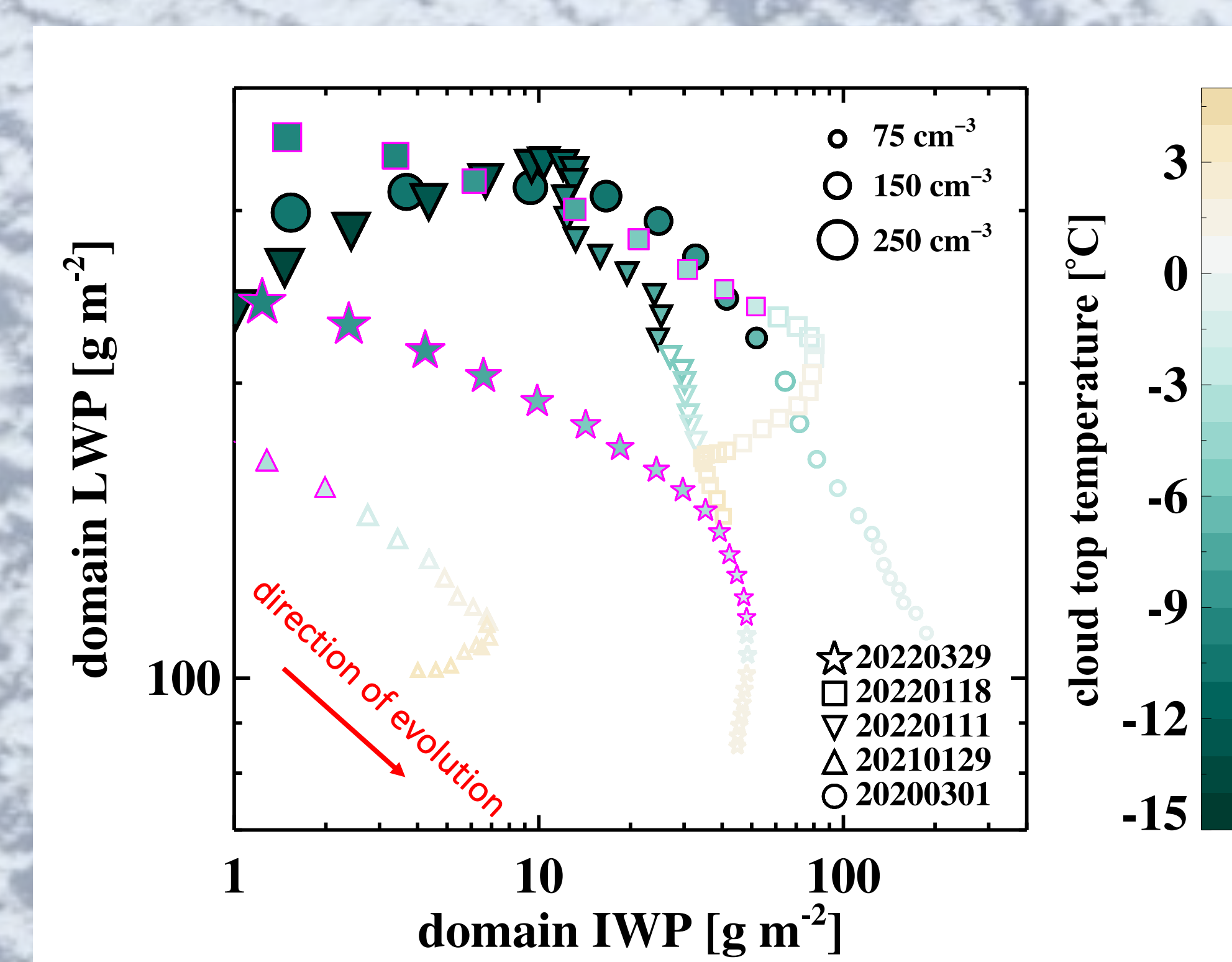
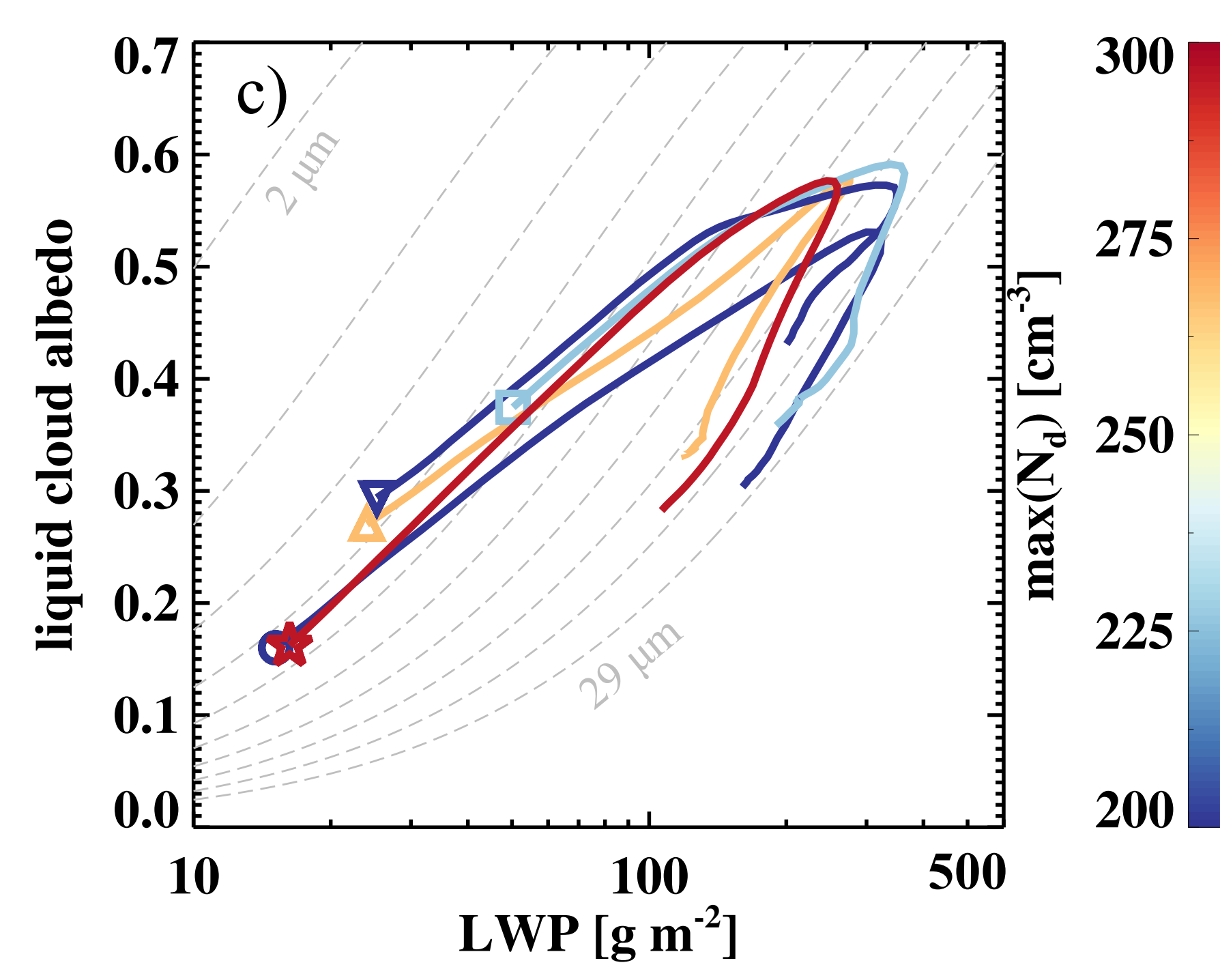
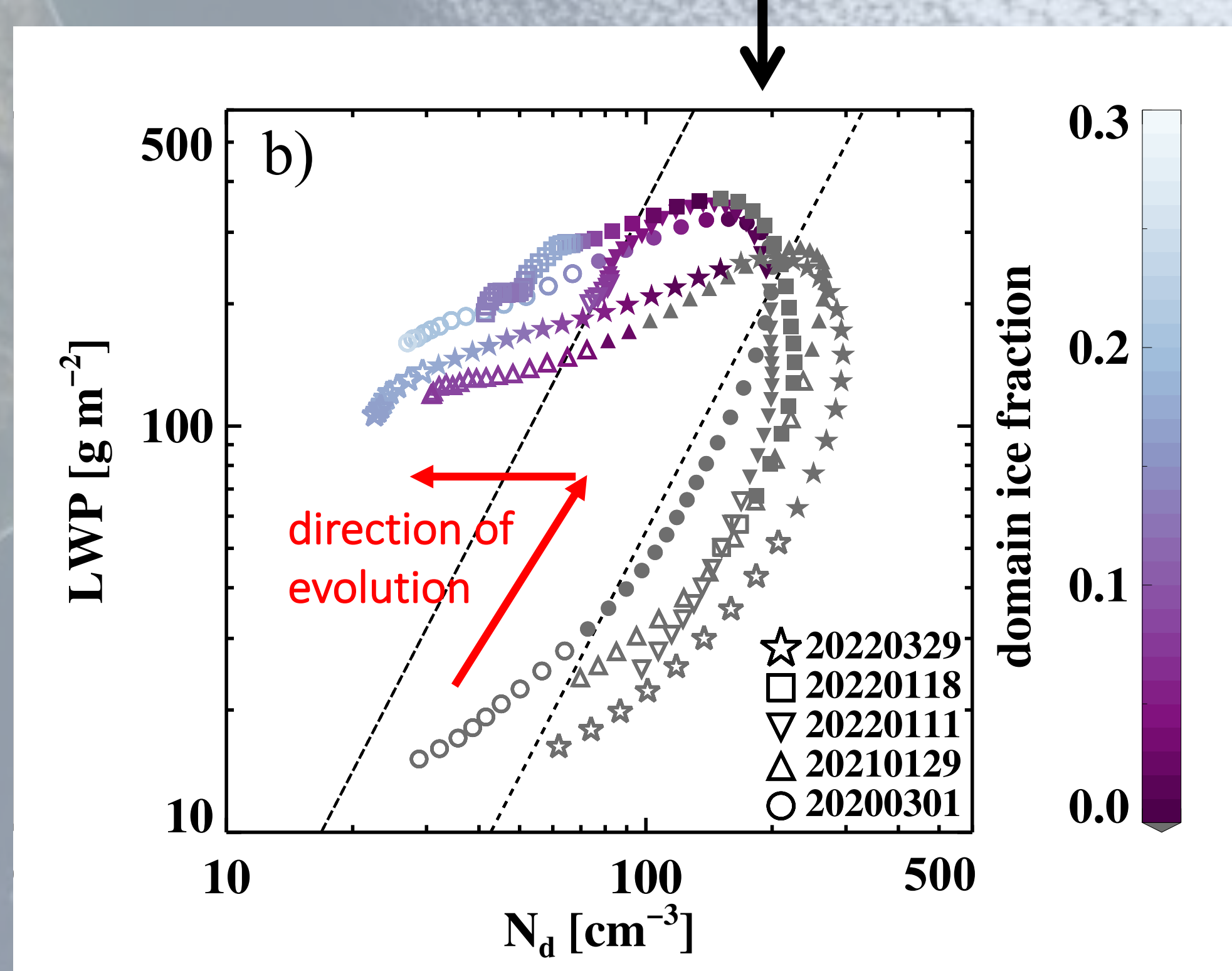
Methods:

- 'time-space' exchange to generate 'static trajectories' for cloud evolution.
- Combination of Micro-Macro physical spaces to infer cloud processes.

LWP- N_d space

Albedo-LWP space

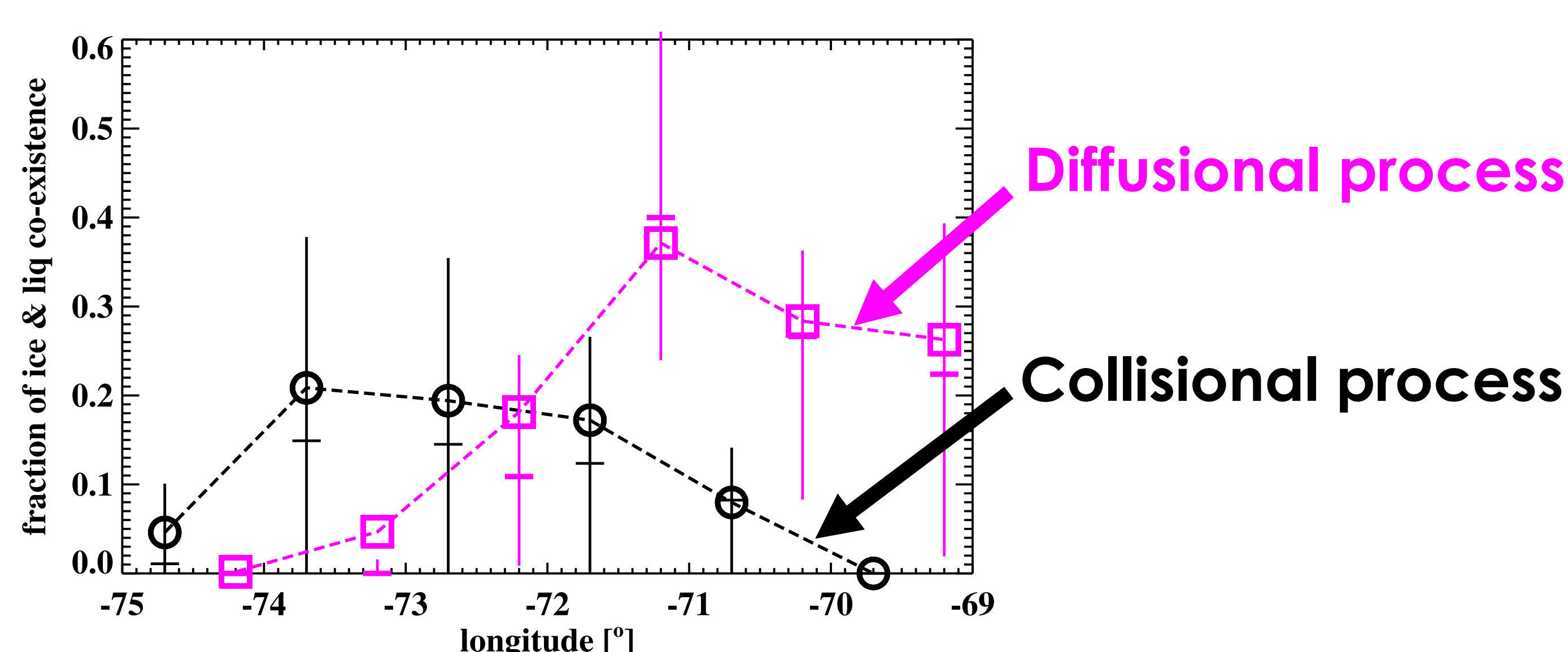
IWP-LWP space



Zhang et al. (2026) Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite, Atmos. Chem. Phys., 26, 6015–6034.



Aircraft validation: ice & liquid co-existence



Cloud organization converges

